

Where the Box Sits: Ax1Gen and Gödel’s Theorem Th3 in S4

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Abstract

Benzmüller and Scott leave open whether Theorem Th3 of their actualist Figure 7—possible existence of a God-like being implies necessary existence, with $P(G)$ derived from the generalized conjunction axiom Ax1Gen—is provable in S4. Two Lean 4 theorems decide that question, and they decide it in opposite directions, according to a single scope. Read literally, as in the Archive of Formal Proofs, Ax1Gen tests positivity of a collection Φ at the current world and reads Φ again under a box. That split forces the accessibility relation to be symmetric, so Th3 holds in S4 with no reflexivity or transitivity hypothesis. Two repairs that put the scopes back together—positivity inside the box, or a world-invariant Φ —still yield the lemma $P(G)$, do not force symmetry, and fail on a two-world S4 chain. The open question is settled entirely by where Ax1Gen looks.

1 A footnote, a notebook, and an oops

Around 1970 Kurt Gödel wrote down, in a page and a half of symbols, a modal version of the ontological argument [11]. The first axiom says that the conjunction of two positive properties is positive. A footnote extends the axiom from two summands to any number of them: the conjunction of a collection of positive properties is itself positive. Dana Scott, working from a copy Gödel allowed him to make, rearranged the package [14, 15, 16]. In Scott’s notes the generalized conjunction is replaced by a direct postulate that being God-like is positive. Scott also inserts a conjunct into the definition of essence, requiring that the essential property actually hold of the individual, and he takes necessary existence to be positive. That last change is the one that keeps the notes off the inconsistent reading of the unrepaired manuscript.

Jordan Howard Sobel later showed that Scott’s premises collapse modality: whatever is true is necessarily true [15, 16]. The collapse has since been machine-checked, and so have repairs in the spirit of Anderson and of Fitting [1, 10, 5]. Separately, the passage from “possibly God-like” to “necessarily God-like” on the Scott side needs symmetry of accessibility, the modal axiom B, and does not need the rest of S5 [12]. None of that is the question of this note. Those facts are recorded again in Section 6 only so that they are not mistaken for the two theorems below.

The question comes from a return to the footnote. Christoph Benzmüller and Dana Scott, in the *Monatshefte für Mathematik* [7], and in the matching Archive of Formal Proofs entry [8], study Gödel’s own generalized conjunction as an axiom they call Ax1Gen. Positivity of being God-like is not assumed. It is lemma L, derived from Ax1Gen. Essence keeps Scott’s extra conjunct. Individual quantifiers in the argument are actualist: an existence predicate says who is around at a world. In their Figure 7, Theorem Th3 is

$$[\Diamond(\exists^E x. Gx) \supset \Box(\exists^E y. Gy)].$$

In the S5 embedding the proof uses symmetry of the accessibility relation and nothing else from S5. The theory `GoedelVariantHOML2inS4` repeats the same non-logical axioms over an S4 frame (reflexive and transitive, symmetry not assumed) and leaves Th3 as `oops`, with the

comment “Open problem”. The header of that theory says that the proof of Th3 fails in S4. Figure 8, which repairs the argument by changing necessary inclusion instead of essence, has the same open `oops` in `GoedelVariantHOML3inS4`. The authors’ formalization reports neither an S4 proof nor an S4 countermodel.

This note records two theorems already checked in Lean 4, in the repository described in the acknowledgement. They answer the open problem for the literal AFP axiom, and they answer it again after two repairs of a scope that the footnote does not ask for. The literal reading proves Th3. The repairs refute it in S4. Which answer one gets is exactly the source/successor split in `Ax1Gen`.

2 Figure 7, as formalized

Worlds form a type W , individuals a type Ind , and accessibility is a relation R on W . A world-proposition is a map $W \rightarrow \mathbf{Prop}$. A property is a map $Ind \rightarrow W \rightarrow \mathbf{Prop}$, individual first and world second, matching the AFP type $e \Rightarrow i \Rightarrow \mathbf{bool}$. Positivity P is a world-relative predicate on properties. Necessity and possibility are the usual quantifications along R :

$$(\Box\varphi)w \equiv \forall v. R w v \rightarrow \varphi v, \quad (\Diamond\varphi)w \equiv \exists v. R w v \wedge \varphi v.$$

Validity $\llbracket\varphi\rrbracket$ means φ holds at every world.

An existence predicate $ex : Ind \rightarrow W \rightarrow \mathbf{Prop}$ plays the role of the AFP constant *existsAt*. The outer type of individuals is fixed. Who exists may vary. Actualist quantifiers are the restrictions

$$(\forall^E y. \varphi y)w \equiv \forall y. ex y w \rightarrow \varphi y w, \quad (\exists^E y. \varphi y)w \equiv \exists y. ex y w \wedge \varphi y w.$$

Quantifiers over properties stay possibilist: they range over every map $Ind \rightarrow W \rightarrow \mathbf{Prop}$.

God-likeness, necessary inclusion, essence, and necessary existence are the Figure 7 clauses. Writing $\varphi \cdot \psi$ for the pointwise conjunction of properties and $\sim \varphi$ for pointwise negation:

$$\begin{aligned} Gx &\equiv \forall\varphi. P\varphi \supset \varphi x, \\ \varphi \supset_N \psi &\equiv \Box(\forall^E y. \varphi y \supset \psi y), \\ \varphi \text{Ess. } x &\equiv \varphi x \wedge (\forall\psi. \psi x \supset (\varphi \supset_N \psi)), \\ Ex &\equiv \forall\varphi. (\varphi \text{Ess. } x) \supset \Box(\exists^E z. \varphi z). \end{aligned}$$

The AFP binder in the last clause shadows x ; the Lean definition uses a fresh variable, which is the same formula. The axioms, all under validity, are `Ax1` (binary conjunction), `Ax2a` (exclusive disjunction of $P\varphi$ and $P(\sim\varphi)$), `Ax2b` ($P\varphi \supset \Box P\varphi$), `Ax3` (PE), and `Ax4` (a positive property passes positivity along $\supset_N$). `Ax1` is part of the package and holds in the one-world model of Section 4. Th3 does not use it. The AFP proof of Th3 does not use it either.

The footnote becomes two abbreviations and one axiom.

$$\begin{aligned} \text{PosProps } \Phi &\equiv \forall\varphi. \Phi\varphi \supset P\varphi, \\ \text{ConjOfPropsFrom } \varphi \Phi &\equiv \Box(\forall^E z. \varphi z \leftrightarrow (\forall\psi. \Phi\psi \supset \psi z)), \\ \text{Ax1Gen} &\equiv \llbracket (\text{PosProps } \Phi \wedge \text{ConjOfPropsFrom } \varphi \Phi) \supset P\varphi \rrbracket. \end{aligned}$$

Both Φ and φ are universally quantified. This is the text of `GoedelVariantHOML2` and of `GoedelVariantHOML2inS4`. Lemma L is $\llbracket PG \rrbracket$, obtained by feeding `Ax1Gen` the collection of properties that are positive at the world of evaluation. It is a lemma, not an axiom. Scott’s later A3 postulates that same sentence outright [14]. The two packages are different, and Section 6 is there to keep them apart.

In the S5 theory, Th3 is proved from Th2 by the three steps the AFP script writes down: existence of a God-like being implies necessary existence; the diamond of the antecedent is therefore a diamond of a box; symmetry turns that diamond-box back into a box. The Lean theorem `th3_of_symmetric` is that argument, with the witness individual written out. Reflexivity is not used.

3 What the encoding is

The development is a shallow embedding in Lean 4 [9], with no Mathlib library. Hypotheses are propositions, passed as arguments. There is no Lean `axiom` declaration anywhere in the repository. The host logic is the ordinary Lean kernel. Where a proof uses classical reasoning, `#print axioms` says so (Section 7). Isabelle/HOL is classical [13], so that use of classical logic matches the host logic of the AFP entry. It is still a different host.

The departures from the AFP text are these.

AFP	This Lean development
Isabelle/HOL bool	Lean <code>Prop</code> . Classical steps are exactly the ones whose axiom list contains <code>Classical.choice</code> .
$[\psi] \equiv \forall w. \psi w$	The definition <code>valid</code> .
$existsAt : e \Rightarrow \sigma$	$ex : Ind \rightarrow W \rightarrow Prop$. A fixed outer type, with existence varying by world, is how the AFP file represents varying domains.
Property quantifiers are HOL $\forall$ over functions $e \Rightarrow \sigma$	Lean $\forall$ over functions $Ind \rightarrow W \rightarrow Prop$. Comprehension is full, as in their impredicative <code>Ax1Gen</code> .
Figure 8 writes $\varphi \neq (\lambda x. \perp)$, which is extensional inequality of HOL functions	<code>neBot</code> : $\neg \forall x w. \varphi x w \leftrightarrow \text{False}$. The symmetry proof supplies a pointwise witness. Lean’s intensional $\varphi \neq \perp$ is a different statement, and it is not the one used.
Deep surface syntax, Leibniz equality inside the object logic, Nitpick	Absent. The one-world model of Section 4 is the cardinality-one shape Nitpick reported, rebuilt by hand.

World-relative P , actualist individual quantifiers, possibilist quantifiers over properties, Figure 7 essence with the conjunct φx , and the derivation of $P(G)$ are the AFP reading. They are not extra deviations. Older modules in the same repository use a different argument order for properties and a constant domain. Those modules are not the encoding of this paper.

One consequence of Lean’s `Prop` is worth saying before the proofs. A proof of a negation may be a classical contradiction rather than a construction of a counter-witness. The symmetry argument is of that kind. The countermodel of Section 5 is not: failure of Th3 at the source is an explicit world, an explicit individual, and a computation.

4 Theorem 1: the literal axiom forces a back-edge

4.1 The trick, before the symbols

`Ax1Gen` is an invitation to define a property by saying which collection it conjoins. The abbreviation does not require the collection to be the same collection at every world. `PosProps` Φ is evaluated at the world where the axiom is applied. `ConjOfPropsFrom` puts Φ under a box, so on the right-hand side Φ is read at successor worlds.

That is enough room for a prank. At the current world, let Φ mark nothing. Positivity of “every marked property” is then vacuous, and the axiom will still conclude that the defined property is positive here. Next door, let Φ mark the empty property, and define the property to be “this world is the one I started from”. The biconditional under the box is happy in both

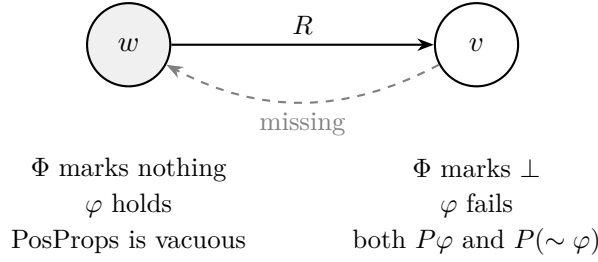


Figure 1: The collection used by `fig7_implies_symmetric`. At the source, Φ is empty, so Ax1Gen owes nobody a positivity premise. At a successor that cannot see the source, the same Φ marks the empty property, φ is “the world is the source”, and Ax4 with Ax2a disagree.

places: at home, an empty collection conjoins to a tautology on the world-coordinate; next door, a collection that contains a contradiction conjoins to a contradiction, and “this is home” is false next door. Positivity, once granted at home, is pushed along the edge by Ax2b. At the far end, “this world is home” necessary-implies its own negation, because the far world cannot see home—that is the edge we are trying to forbid—so the inclusion is vacuous. Ax4 passes positivity to the negation. Ax2a refuses to let a property and its negation both be positive.

The frame never had to be reflexive or transitive. Any edge without a back-edge is unsatisfiable for Ax1Gen, Ax2a, Ax2b, and Ax4. S4 is the special case in which one hoped to do without the back-edge.

4.2 The argument

Theorem 1 (`fig7_implies_symmetric`). *From Ax1Gen, Ax2a, Ax2b, and Ax4 it follows that R is symmetric. Reflexivity and transitivity are not hypotheses.*

Sketch. Fix $R w v$, and suppose $\neg R v w$. Define a property and a collection by

$$\begin{aligned}\varphi &::= \lambda _ u. (u = w), \\ \Phi &::= \lambda \psi u. (u \neq w \wedge \forall x. \neg \psi x u).\end{aligned}$$

At w , the clause $u \neq w$ fails, so Φ marks no property and $\text{PosProps } \Phi$ holds outright.

The boxed biconditional holds at every successor u of w . If $u = w$, the right-hand side is vacuous and the left-hand side says that the world is w . If $u \neq w$, then Φ marks $\perp$. The right-hand side asks $\perp$ of an individual and is impossible, while the left-hand side, “the world is w ”, is false. Both sides fail together. (The impossibility is a classical contradiction: if the world were not w , $\perp$ would be marked and could not hold.)

Ax1Gen therefore yields $P\varphi$ at w . Ax2b pushes it to v . At v , necessary inclusion of φ in $\sim\varphi$ is vacuous. A successor of v cannot be w , because v does not see w , so the antecedent φ never fires. Ax4 yields $P(\sim\varphi)$ at v . Ax2a says that $P\varphi$ and $P(\sim\varphi)$ are exclusive. Contradiction. Hence $R v w$. $\square$

Ax3 is not used. The same four hypotheses are therefore unsatisfiable on every non-symmetric frame. In particular they are unsatisfiable on the two-world chain of Figure 2, which is the content of `fig7_unsat_on_S4_chain`. That chain is an S4 frame. It is not a countermodel to literal Figure 7, because no positivity and no existence predicate satisfy the axioms there.

Theorem 2 (`th3_fig7`). *From Ax1Gen, Ax2a, Ax2b, Ax3, and Ax4,*

$$[\Diamond(\exists^E x. Gx) \supset \Box(\exists^E y. Gy)].$$

There is no frame hypothesis.

Sketch. Theorem 1 supplies symmetry. The AFP argument then runs, as `th3_of_symmetric`. A diamond at w is a successor v at which some existing individual is God-like. Theorem Th2, which is $Gx \supset \Box(\exists^E G)$ and which needs Ax2a, Ax2b, and Ax3, places a God-like being at every successor of v . Symmetry puts w among them. Th2 at w is the box required by Th3. $\square$

The intermediate AFP lemmas that do not mention symmetry are in the same file, and they are rediscoveries of the AFP proofs: `lemma_L`, the forward spread of negative positivity `ax2b'`, `th1_fig7` (God-likeness is an essence), and `th2_fig7`. Th4, possible existence, and Th5, necessary existence at every world, are likewise rediscoveries for Figure 7. Th4 is the usual indirect argument: if G were nowhere exemplified along R , then G would necessarily include $\perp$, Ax4 would make $\perp$ positive, and Ax2a would refuse. The one-world control below shows that the axioms are not jointly absurd once a back-edge is allowed. On a one-point frame the back-edge is the reflexive loop, so S4 and S5 agree there. The interesting content of Theorem 2 is the case where one tries to withhold it.

Proposition 3 (`unit_fig7_th3`). *On a single world, with the identity relation, one individual existing there, and principal positivity $P\varphi \equiv \varphi()()$, axioms Ax1, Ax2a, Ax2b, Ax3, Ax4, and Ax1Gen hold, and Th3 holds.*

This is the shape Nitpick reported at cardinality one, checked directly. It is a consistency witness for the literal package. It is not a claim that the axioms are true of anything outside the model.

4.3 Figure 8, and what is not claimed

Figure 8 (`GoedelVariantHOML3`) changes two clauses. Necessary inclusion gains a side condition, and essence loses the conjunct φx :

$$\varphi \supset_N \psi \equiv \Box(\varphi \neq (\lambda x. \perp) \wedge \forall^E y. \varphi y \supset \psi y),$$

with essence the universal closure of that inclusion under properties the individual has. In the Lean file the inequality is `neBot`, the extensional disagreement of Section 3. The S4 theory again leaves Th3 as `oops`, comment “Open problem”.

Theorem 4 (`th3_fig8`). *From literal Ax1Gen, Ax2a, Ax2b, Figure 8’s Ax3, and Figure 8’s Ax4, Th3 holds, with no frame hypothesis.*

The symmetry argument is the one in Theorem 1. Figure 8’s inclusion asks, in addition, that φ not be identically false. The witness property “the world is w ” disagrees with $\perp$ at w , once any individual is on hand to point at. Theorem `th3_fig8` takes that individual from the diamond in the antecedent of Th3, proves symmetry from it, and only then runs the back-edge. If the type of individuals were empty the diamond could not hold, so the implication would be idle. The companion `fig8_implies_symmetric` states the extra inhabitant as a hypothesis, because symmetry alone, with no diamond, has to be told that someone exists. On the S4 chain, once an individual is given, the Figure 8 axioms used for symmetry are unsatisfiable (`fig8_unsat_on_S4_chain`).

Figure 8’s Th4 is not claimed, and neither is a Figure 8 Th5. The Figure 7 proof that G is possibly exemplified sends G down to $\perp$ and then sends $\perp$ up to $\sim \perp$. For Figure 7 the second inclusion is vacuous: a false antecedent implies anything. Figure 8’s inclusion requires the antecedent property to be distinct from $\perp$, and $\perp$ is not. In `GoedelVariantHOML3.thy` the Th4 script is `oops`, and Th4 is then reintroduced by `axiomatization`. Copying that postulate into Lean would not be a proof. It is not one here.

4.4 Why Theorem 1 may be an accident of the abbreviation

Theorem 2 is a theorem of the literal AFP axiom. It may also be a theorem one did not mean to state.

Gödel’s footnote, as the generalized conjunction is used in this literature, says that a conjunction of *positive* properties is positive [11, 4]. In the abbreviation, “the conjuncts are positive” is judged at the source, and “ φ is their conjunction” is judged at successor worlds, where Φ is evaluated afresh. The collection in Figure 1 is empty at home and full of contradictions next door. Nothing in the footnote asks a collection to change its mind on the way out of the world. The split is what a shallow embedding does if PosProps is left outside a box while the predicate Φ still depends on the world.

That is a feature of the definition as written in `GoedelVariantHOML2`. It is a fair question whether it is a feature anyone wanted. Theorem 2 would be a strange place to stop. The next section is the reason for not stopping: two repairs that a reader of the footnote might have thought were already the axiom. Under either repair, lemma L survives, symmetry is not forced, and Th3 fails in S4.

5 Theorem 2: repair the scope, and S4 wins

Everything in Figure 7 stays put except Ax1Gen. There are two replacements. They are not rivals for a single idea.

R1, positivity in the box. The closer repair judges both halves at the same accessible worlds:

$$[\Box(\text{PosProps } \Phi \wedge \forall^E z. \varphi z \leftrightarrow (\forall \psi. \Phi \psi \supset \psi z)) \supset P \varphi].$$

Call this `Ax1GenInBox`. In K, a box of a conjunction is a conjunction of boxes, and `ConjOfPropsFrom` is already a box, so the displayed form is equivalent to

$$[(\Box(\text{PosProps } \Phi) \wedge \text{ConjOfPropsFrom } \varphi \Phi) \supset P \varphi],$$

which is `Ax1GenBox`. The equivalence `ax1GenBox_iff_inBox` depends on no kernel axioms. One countermodel covers both spellings. This is the reading on which “positive” and “conjunction” are said of the same world.

R2, a rigid collection. Restrict Ax1Gen to collections that do not depend on the world: $\Phi \psi w \leftrightarrow \Phi \psi v$ for every property and every pair of worlds. PosProps and ConjOfPropsFrom stay literal. Call this `Ax1GenRigid`. The prank collection of Figure 1 is not rigid, so R2 never sees it. R2 is a different restriction, not a second spelling of R1.

5.1 Lemma L still derives

Under R1, lemma L is immediate from the repaired axiom alone (`lemma_L_box`, `lemma_L_inBox`). Take Φ to be P itself. At every world, every property that Φ marks is positive there, by identity, so the extra box is free. The conjunction of whatever is positive at a world is G at that world, by the definition of G .

Under R2 the same Φ may be refused, because P itself need not be rigid. The repair is to take a snapshot at the world w where the lemma is being proved: $\Phi \psi _ \equiv P \psi w$. That collection ignores its world argument, so it is rigid. At a successor v , the conjunction it defines is G only if positivity at w and positivity at v agree. Ax2a and Ax2b give exactly that agreement along an edge (`pos_agree`): a positive property stays positive by Ax2b, and a property positive at v cannot have a positive negation at w , or Ax2b would carry the negation forward and break Ax2a at v . So `lemma_L_rigid` needs Ax2a and Ax2b. Literal `lemma_L` needed neither. The box

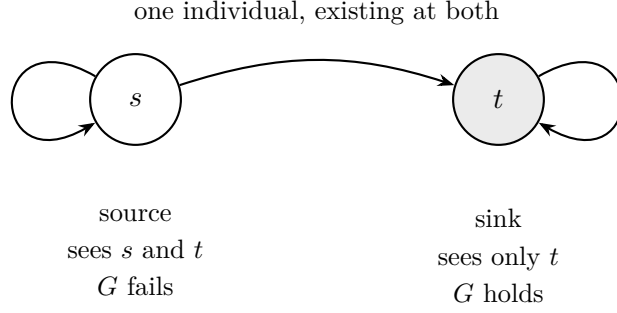


Figure 2: The S4 chain of R_chain , drawn as source $s = \text{false}$ and sink $t = \text{true}$. Reflexive, transitive, and not symmetric. Under $chainP$, a God-like being exists at the sink only, so at the source a God-like being is possible and not necessary. Literal $Ax1Gen$ has no model on this frame. Both repairs do.

in $R1$ hides the same fact, because $\Phi := P$ is re-read at the successor rather than frozen at the source.

Symmetry is not a consequence of either repair. The collection in Theorem 1 is empty at the source, so source PosProps holds, but at a successor it marks $\perp$. $R1$'s boxed PosProps demands $P(\perp)$ there. $R2$ does not apply, because the collection varies. The four-hypothesis derivation of a back-edge has nothing left to instantiate.

5.2 The chain

Let the worlds be the two Booleans. The source false sees both worlds. The sink true sees only itself (Figure 2). The relation is reflexive and transitive. It is not symmetric: the sink does not see the source. One individual exists at both worlds. Positivity ignores the world at which it is asked and reads the sink:

$$P\varphi \equiv \varphi() \text{true}.$$

A property is positive precisely when the individual has it at t .

At the sink, God-likeness is a tautology. G says that every positive property holds of the individual there, and positivity means that the property holds of the individual there ($chain_god_sink$; no kernel axioms). At the source it fails. The property “the world is the sink” holds at the sink, hence is positive, and the individual does not have it at the source ($chain_not_god_source$; no kernel axioms).

Theorem 5 ($chain_th3_fails_at_source$). *At the source, $\Diamond(\exists^E G)$ holds and $\Box(\exists^E G)$ fails.*

Sketch. The source sees the sink, where the individual exists and is God-like. The source also sees itself, where that same individual exists and is not God-like. The diamond is the first fact. The failure of the box is the second. $\square$

No kernel axiom is used. The packaged denial $chain_not_Th3$ is the same fact pulled back along the definition of $Th3$, and it is equally axiom-free.

Theorem 6 ($r1_s4_countermodel$, $r2_s4_countermodel$). *On this frame the relation is an S4 frame and not a symmetric one. $Ax1$, $Ax2a$, $Ax2b$, $Ax3$, $Ax4$, and both spellings of $R1$ hold, and $Th3$ fails. The same frame, the same positivity, and the same existence predicate satisfy $R2$, and $Th3$ fails.*

The axiom checks are direct, and most of them use no kernel axioms. $Ax2a$ is the exclusive-or of “ φ holds at the sink” and “ φ fails at the sink”, which is classical excluded middle ($chain_ax2a$). $Ax2b$ is immediate because P does not look at its world argument: positivity is already global.

Reading of Ax1Gen	Lemma $P(G)$	Symmetry forced	Th3 in S4
Literal AFP	from Ax1Gen alone	yes (Thm. 1)	proved (Thm. 2)
R1, boxed	from R1 alone	no	fails (Thm. 6)
R2, rigid Φ	needs Ax2a and Ax2b	no	fails (Thm. 6)

Table 1: The S4 question, by scope. The witness for both negative cells is Figure 2.

Ax4 follows because both worlds see the sink, so a necessary inclusion can be read there, where P evaluates extensions. Ax3 holds because P on necessary existence only asks for the extension at the sink, and the essence hypothesis at the sink includes that the property holds of the individual there. For R1, the boxed biconditional at whichever world one starts from can be read at the sink, which both worlds see, and the right-hand side at the sink is exactly the extension P consults. For R2 the same computation works, and rigidity lets a mark at the sink be copied back to the world where PosProps was assumed.

Proposition 7 (`chain_not_literal_Ax1Gen`). *This positivity does not satisfy literal Ax1Gen on the chain.*

If it did, Theorem 1 would make the chain symmetric, and `fig7_unsat_on_S4_chain` already records that contradiction. The repairs are strictly weaker than the literal axiom on this frame. Theorem 2 is not retracted. It is fenced. A model of R1 or of R2 need not be a model of the abbreviation that proved it.

Table 1 is the whole comparison. The possibilist twin `GoedelVariantHOML2possInS4` also leaves Th3 as `oops`. It is not encoded. A constant domain with no existence predicate would be a different theorem, and the Scott-style chain already studied in this repository is that different theorem (Section 6).

6 What was already known

Several nearby statements are in the same repository, and they were already theorems of other people. They are named here so that the two results above are not asked to carry them.

Sobel showed that Scott-style premises yield $\varphi \supset \Box\varphi$ [15, 16]. Benzmüller and Fuenmayor encoded Scott’s, Anderson’s, and Fitting’s variants and checked that Scott’s intensional positivity collapses modality, while Anderson’s emendation and Fitting’s extensional positivity do not [5, 1, 2, 10]. The repository’s collapse modules rediscover the Scott-side collapse, including the world-relative variant in which collapse along R is separated from Sobel’s global biconditional when R is not universal. That is a rediscovery. It is not Theorem 2 or Theorem 6.

Kanckos and Woltzenlogel Paleo showed that Scott’s argument goes through in KB: symmetry suffices, and S5 is more than the proof uses [12]. Benzmüller and Scott make the same point for their Th3, which cites R_{symm} and not reflexivity or transitivity [7, 8]. The repository’s Scott-style development rediscover the positive half (local Th3 from symmetry alone, on that encoding) and pairs it with countermodels once symmetry is dropped. Those countermodels postulate positivity of being God-like. Figure 7 derives that sentence as lemma L. A countermodel that assumes the lemma does not answer a question whose point is what happens when the lemma is earned from Ax1Gen. The file `S4open.lean` in the repository is an explicit record of this mismatch. Theorem 6 is not that countermodel. On the chain of Figure 2 the literal axiom has no model at all.

Anderson’s emendation—only one direction of the exclusion axiom, and God-likeness as possession of exactly the positive properties, necessarily—is a known repair [1, 2]. A thin fragment of it is formalized elsewhere in the repository, and a two-world witness there keeps a contingent fact alive. That fragment is a rediscovery of Anderson’s repair. Fitting’s extensional block is not formalized [10]. Neither is part of Theorems 2 and 6.

The early mechanizations of the Scott-style argument [6] are the same Scott package. Benzmüller’s note on ultrafilters [4] derives $\Box(\exists^E x. Gx)$ already in K, from an ultrafilter axiom U1, a conjunction principle, an entailment axiom, and the definition of G . The note is explicit that this is a simplified package, and that Scott’s own A3 postulated $P(G)$ in place of the footnote’s conjunction. It is not Figure 7’s axiom list, and it does not use the AFP scoping of Ax1Gen. The Archive entry on simplified variants [3] studies arguments that avoid essence and necessary existence and that are valid in K or KT. Those are different axioms again.

What was checked, for the claim that these two theorems were not found already, is the following AFP theories, together with the S4 and S5 embeddings they import:

- `GoedelVariantHOML2` and `GoedelVariantHOML2inS4` (Figure 7, S5 and S4);
- `GoedelVariantHOML3` and `GoedelVariantHOML3inS4` (Figure 8, S5 and S4);
- `GoedelVariantHOML2possInS4` (the possibilist S4 twin, not encoded here).

The literature named in this section was checked as well, namely the items cited from the repository notes that accompany the formalization: Sobel [15, 16], Anderson [1], Anderson and Gettings [2], Fitting [10], Kanckos and Woltzenlogel Paleo [12], Benzmüller and Woltzenlogel Paleo [6], Benzmüller and Fuenmayor [5], Benzmüller [4], and the simplified-argument AFP entry [3]. In the AFP S4 files, Th3 is the open `oops`. In [4] the K-theorem is the ultrafilter package just described. No proof or countermodel of literal Figure 7 Th3 in S4, and no countermodel of the two repaired readings, turned up in those sources. That is a statement about where the search went. It is not a priority claim.

7 What the kernel accepts

The development builds with `lake build` on Lean 4.34.0, toolchain `leanprover/lean4:v4.34.0`, in 35 jobs, with no Mathlib. A search of the Lean sources finds no `sorry`, no `native_decide`, and no `axiom` command. The strings that look like those words sit in comments. Hypotheses of the ontological argument are parameters of type `Prop`.

Table 2 is the output of `#print axioms` on the headline theorems, run in that toolchain after the build. “None” means Lean reported that the constant does not depend on any axioms. The triple `propext`, `Classical.choice`, `Quot.sound` is the standard classical package of the Lean kernel: propositional extensionality, choice, and the soundness of quotients that the classical development uses. `Classical.choice` enters through `Classical.byContradiction` in the symmetry argument and in Figure 7’s Th4, and through `Classical.em` in the one-world check of Ax2a and in `chain_ax2a`. `propext` on its own is the chain’s transitivity and the proof that the chain is not symmetric, both of which simplify propositional equalities. No custom axiom appears.

8 The moral

Theorem 2 says that the literal Figure 7 axioms prove Th3, and prove it in K, hence in S4. The mechanism is the empty-at-home collection of Figure 1, which the AFP abbreviation permits because positivity is read at the source and the collection is read again under the box. Theorem 6 says that if positivity moves inside the box, or if the collection is required to be the same collection at every world, then lemma L still derives and the two-world chain is a countermodel to Th3 in S4. The chain does not satisfy the literal axiom. The S4 question, for this axiom list, is decided by that scope and by nothing else in the modal logic.

Theorem	#print axioms
lemma_L, ax2b', th1_fig7, th2_fig7, th3_of_symmetric	none
th1_fig8, th2_fig8, th3_fig8_of_symmetric	none
fig7_implies_symmetric, th3_fig7, fig7_unsat_on_S4_chain	propext, Classical.choice, Quot.sound
th3_fig8, fig8_implies_symmetric, fig8_unsat_on_S4_chain	propext, Classical.choice, Quot.sound
unit_fig7_th3	propext, Classical.choice, Quot.sound
ax1GenBox_iff_inBox	none
lemma_L_box, lemma_L_inBox, lemma_L_rigid	none
chain_th3_fails_at_source	none
chain_not_literal_Ax1Gen	propext, Classical.choice, Quot.sound
r1_s4_countermodel, r2_s4_countermodel	propext, Classical.choice, Quot.sound

Table 2: Kernel axioms of the headline theorems, Lean 4.34.0. The repaired countermodels inherit `propext` from the chain’s frame facts and `Classical.choice` from `Ax2a`. The failure of `Th3` at the source, taken by itself, depends on none of them.

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